

Chapter 9

Incorporating Industry 4.0 in Corporate Strategy

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ABSTRACT

Knowledge tells us that increase in productivity decreases margins in a competitive market and increases margins in a growing market. Not much is known about Industry 4.0 and its position within the corporate strategy and its possible impact on corporate performance. This chapter discusses the position of Industry 4.0 within the corporate strategy and how it may impact corporate performance. The main points include that corporates have to reflect on their core resources, leadership, and knowledge portfolio to take advantage of the Industry 4.0 platform. The corporates engaging in cost and volume strategies and those engaging in fast new product development strategies may benefit greatly with Industry 4.0. The services of Industry 4.0 may be outsourced but with added risks. Finally, the increased productivity with Industry 4.0 under constraint market growth may lead to the potential risk of market failure and large-scale layoffs.

INTRODUCTION

The primary mission of any corporation is to maximize profitability on behalf of its shareholders (Friedman, 2009) and to provide the best value to its stakeholders (Freeman, 1984). The development of technologies such as the big data and data analysis, IoT, artificial intelligence (AI), flexible robotics, 3D printing, augmented reality, 3D holographic scanning smarter sensors, greater miniaturization, cloud computing, customer feedback and Customer management software (CMS), location detection technologies such as radio-frequency identification (RFID), natural language programming, and chang-

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ing global social-political environments are all driving human productivity toward a new industrial revolution (Herter & Ovtcharova, 2016; Reinhard, Jesper, & Stefan, 2016). These new technologies are consequently increasing the productivity of the corporations and changing the market equilibrium. The more these technologies evolve and are accepted by the industry, consequently, the industrial productivity shall proportionately increase. However, as the productivity of the corporation increases, it may or may not proportionately increase the performance and profitability.

Contrary to popular understanding, any impulsive investment in technology may increase the corporate debt and, consequently, decrease the corporate performance. Technology adoption and subsequent productivity change may also distort the market equilibrium, which, will force the competitors to engage in similar cost/volume strategies, lowering the overall margins per unit of production (Porter, 1979) for the whole industry. With these possible consequences in mind, it is important to logically weigh where in the value chain would the Industry 4.0 platform technology increase the value for the corporation and how these technologies would contribute to the competitiveness in the long run.

In the background of the evolving Industry 4.0 technological environments, this chapter provides a framework on how the corporations can increase their performance by reflecting on the role of Industry 4.0 on the overall corporate strategy. While most of the literature on Industry 4.0 is on automation (Mrugalska & Wyrwicka, 2017), productivity, benefits, and business models (Man & Strandhagen, 2017), this chapter tries to create a dialogue between academics of the Industry 4.0 and those who study the corporate strategy and corporate performance. A clearer understanding of Industry 4.0 and its position in corporate strategy will help us in understanding better how to develop business models and strategize the “may or buy” decision on Industry 4.0 technology.

The chapter is structured in the following way: First, it briefly reviews Industry 4.0. Next, this is followed by an assessment of corporate strategy and strategy landscape. Finally, the chapter discusses the position of Industry 4.0 in corporate strategy landscape and how and where it brings value to corporate performance. The chapter presents eleven propositions that give strategy-level perspectives on Industry 4.0.

LITERATURE REVIEW

The motivation of this chapter is to provide a slightly different perspective on Industry 4.0 where the authors contextualize it in the corporate strategy. Most of the scholarships around Industry 4.0 are on the disruptive increase in productivity, automation, new market alpha, and industry digitalization, but the research rarely makes any conversation on corporate strategy. It is in this context that this chapter adds value.

A BRIEF OVERVIEW OF INDUSTRY 4.0

Industrial revolutions have always been accompanied with a disruptive increase in productivity. The increase in productivity has also increased wages and, subsequently, the quality of life of the citizens. The evolution of Industry 4.0 is incremental, yet its effect will have a disruptive impact on corporate performance, productivity and overall markets and industries. One can understand it better when studying all the different stages of industrial revolutions (Figure 1).

Industry 1.0 was the first generation of industrial revolution that started around late 18th century. During this generation, it was the first time that machines, instead of animals, started to support humans in manufacturing and production. The invention of water turbines and steam engines helped the people-machine interface by increasing their productivity substantially. The factories owned the energy-generation systems. The management structures of these manufacturing and production units were highly pyramidal and feudal. The productivity was higher than before, and the degree of human intervention was very high.

Industry 2.0 evolved from the first generation after the introduction of electricity in the 1890s. The electrical energy-driven machines further changed the outlook of the industry. It increased industrial productivity. In this era, the productivity was higher and easier, leading more corporations to engage in production and manufacturing activities. With limited market capacity, this led to increased competition. The competition also led to a faster innovation, resulting in the division of labor and the structuring of the study of business and management into a more specialized field. This was also the era, when the European firms were aggressively looking for newer markets especially in third world countries. The theories of scientific management were formulated during this time. This was also the period when business schools started to emerge and management as a field of inquiry started to take structure.

Industry 3.0 evolved from the second generation of industrial revolution. It started around the 1970s when information technology (IT) integrated management systems were introduced. IT significantly reduced transaction costs and control costs of managing and running a corporation. IT-based management systems also enabled the decoupling of organizations' core strength from manufacturing. Post Industry 3.0, the organizational core competency and core value proposition were decoupled from manufacturing and production. For example, Nike, Zara, General Motors, Apple, and Dell are globally renowned for their products. These companies own the brand, research and development (R&D), and the intellectual rights of the products but they do not own the manufacturing. The generation of Industry 3.0 enabled production as a service. This was one major change over Industries 1.0 and 2.0. The organizational design was divided into more specialized functions such as marketing, finance, manufacturing, corporate Entrepreneurship, R&D, and IT. Management research became more scientific and started using more advanced methods for innovation. Consequently, management study was further classified into more specialized fields of study.

Industry 4.0 is an incremental innovation over Industry 3.0 but with the integration of AI and IoT, the disruptive impact on productivity will be far greater than the previous generations of revolution. The term was first used among German manufacturing associations and corporations who were the first ones to rapidly adopt the Industry 4.0 technologies. Industry 4.0 platform is a German government-funded think tank that disseminates information on Industry 4.0. According to them, Industry 4.0 represents a further integration of computing, IT, robotics, Internet, big data, blockchain, and decentralized independent AI based machines which can take corrective decisions while manufacturing. In the world of Industry 4.0, people, machines, equipment, logistics systems, and products communicate and cooperate with each other directly. Production and logistics processes are integrated intelligently across company boundaries to make manufacturing more efficient and flexible (Schweichhart, 2016). According to the McKinsey and Company, the Industry 4.0 is the integration of the manufacturing sector with the IoT framework using the latest sensors, integrating production systems with the back-end supply chain and with the front-end market demand functions using analytic powers, and further reducing contractual risks by using blockchain technologies and the latest manufacturing techniques, eliminating human intervention all together (Wee, Kelly, Cattel, & Breunig, 2015). Management structures during Industry 4.0 are

imagined to be more decentralized and flatter, albeit more specialized. Newer organizational functions such as expertise in data analytics are prominently getting updated into management.

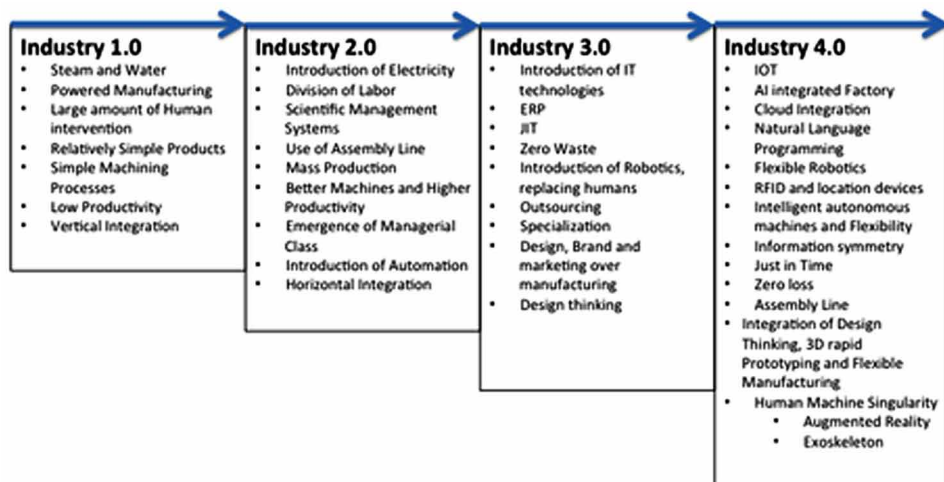
Industry 4.0 is an outcome of the integration of a series of innovative technologies of productivity, creativity, and Industry 3.0 into a single platform. The main scope of Industry 4.0 lies in the reorganization of industrial production methods by incorporating internet of things (IoT), Information and Communications Technology (ICT), big data and AI technologies, robotics, and RFIDs and integrating them with customer feedbacks in order to be (a) more green and sustainable in the industrial processes and supply chains, (b) create a new strategic advantage in the market, (c) reduce costs by using autonomous machines, and (d) ensure faster product development.

For example, initially the RFIDs inside the warehouse were primarily used for tracking the inventory and optimizing warehouse management. However, now the scope of RFID has expanded. They have been tried, tested, and integrated with production systems. This integration has reduced the dependence on assembly line worker and increased the control and monitoring of assembly line production machines. The RFID attached to the work parts are designed to send the problem to the machine and, accordingly, the machine processes the parts into a finished product. The machined parts are then transported to another workstation by autonomous robots, eliminating the need of a fixed assembly line. Earlier, there was a dedicated staff to program and supervise the machines, fix the assembly line, and set the tools in workstations. However, now all these steps are automated, eliminating human interventions. With RFID and autonomous robots, the product design to manufacturing time has been significantly reduced (Whelan, 2016) .

Most research on Industry 4.0 is focused on the RAMI 4.0 framework (Schweichhart, 2016). The conversation is around how to implement Industry 4.0 business models and how these will increase the productivity; however, there is still no discussion on the strategic impact on the corporations post industry 4.0 implementation. Unlike in the previous generations where human intervention was high, in the ideal case, it is assumed that human intervention in the system of production would be completely eliminated. How would the corporations strategize this new trend? What capabilities would the corpora-

Figure 1. Evolution of Industry 4.0

Source: See also (Reinhard, G., Jesper, V., & Stefan, S., 2016; Wee, D., Kelly, R., Cattel, J., & Breunig, M., 2015).



tions require in order to manage this? How would the market change subsequently? It is presumed that the overall strategic impact on corporates would change disruptively post Industry 4.0. This also calls for a deeper study at the social, corporate, and technology levels of the overall impact of Industry 4.0.

CORPORATE STRATEGY AND CORPORATE PERFORMANCE

The Corporate Strategy is a set of guidelines and drivers that direct the decisions and actions of a corporation in order to improve the corporate performance in the short and long run under conditions of uncertainty (Porter, 1980a; Tallman, 1991). There are many corporate strategies such as market strategy, cost strategy, volume strategy, new product strategy, wage arbitrage, new market strategy, and non-market strategy all of which are undertaken to increase a corporations' performance. The corporation engages with all these strategies at multiple levels to enhance the corporate performance. The role of a strategy is not limited to just increasing the revenues or decreasing the input cost. It also includes creating brand value and brand recall—New Market Entry, Credit Rating, ability to attract government subsidy, ability to win government projects such as Lockheed Martin or EADS, and greater market access. Thus, the corporate performance is not narrowly dependent on profitability but also includes wider measures such as profitability, survivability, credit rating, longevity, ability to attract public funding, brand value, and increasing market access and market penetration (Rao, 1994).

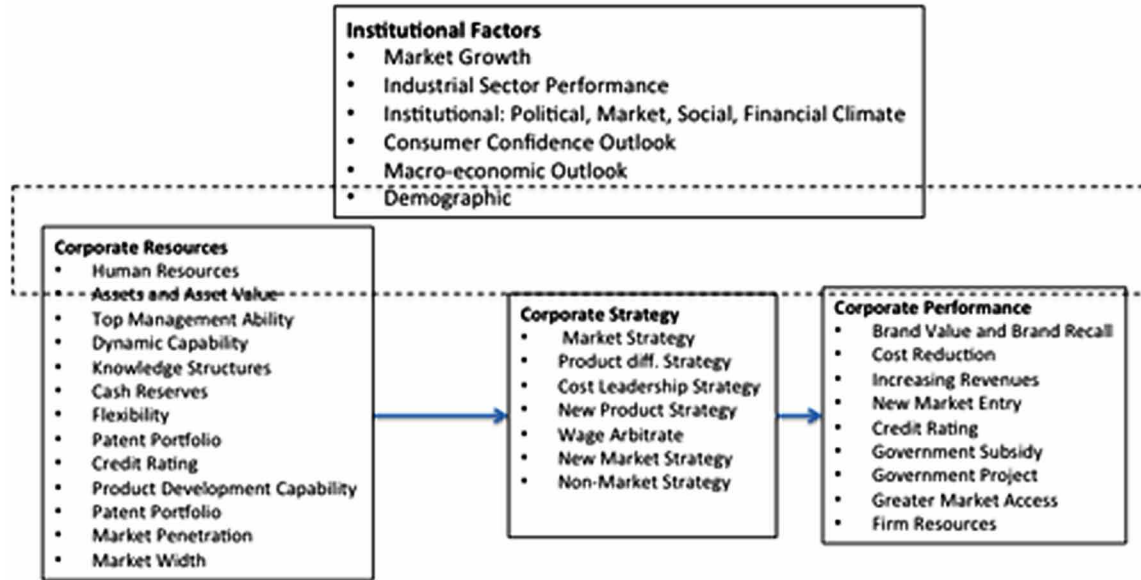
Corporate strategies are driven by two critical antecedents (Grøgaard, 2012). First, the portfolio of resources it possesses forms a major driver of corporate strategy (Morgan, Vorhies, & Schlegelmilch, 2006). Corporate resources include tangible resources such as the reputation of the top management team, learning capabilities, patent portfolio, cash reserves, credit rating, market width and penetration, product development /manufacturing flexibility, and market knowledge. Second, the corporate strategy is developed by environmental characteristics, institutional norms, market cycles, and international and national trade treaties (Freeman, 1984; Tallman, 1991).

Figure 2 represents a framework of corporate strategy and corporate performance. The role of corporate strategy is to take into account the present resources, understand the external market environment and corporate resources, evaluate the market dynamics, and develop strategies to manage the sustainability of corporate performance over short- and long-term in the changing market – industrial environment. As corporations adopt Industry 4.0, how would corporate strategy change and how would corporates strategize in a dynamic competitive environment?

POSITION OF INDUSTRY 4.0 WITH RESPECT TO CORPORATE STRATEGY AND CORPORATE PERFORMANCE

The ability of any corporation to improvise upon its capabilities and strategize tend to help it survive longer (Ruiz-Ortega & García-Villaverde, 2008). Successful corporations such as IBM, GE, and BMW have shown remarkable ability to learn, survive, and grow. On the other hand, corporations (such as Nokia and Kodak), which were market leaders at some point in time but lacked the ability to iterate and strategize based on the changing market and technological environments ended in eventual failure (Binns,

*Figure 2. Antecedents, Actions, and Outcomes of Corporate Strategy
(Strategy Landscape) (Authors' Source)*



Harreld, & Tushman, 2014). While blindly absorbing a new technology may increase the risks of investment failure, failing to timely adopt will also lead to complete irrelevance in the market. Given the high impetus on the promise of Industry 4.0, many corporations are making substantial capital investments to strategize the future. The risk of irrelevance because of the lack of timely adoption outweighs the initial capital expenditure (Sosna, Treviño-Rodríguez, & Velamuri, 2010). Consequently, it is important for the corporations to reflect on their capabilities, their ability to learn and subsequently, prioritize actions accordingly. In this section, the chapter presents how the implementation and absorption of Industry 4.0 are dependent on the corporations's capabilities. Further, the chapter shows how Industry 4.0 supports the corporate strategy and how it positively improves the corporate performance.

Impact of Corporate Resources on Industry 4.0

As has been observed in the previous industrial revolutions, each stage of the revolution brought new organizational designs and the requirement of a new type of a skill set. The complexities of Industry 4.0 would require new organizational capabilities and resources. Organizational resources and capabilities of production are closely interlinked in order to create a long and lasting shareholder value. The capabilities of production have to be holistically aligned with the corporate resources and corporate strategy in order to create sustainable competitive corporate performance (Arthurs & Busenitz, 2006). Before integrating Industry 4.0, corporations have to reflect upon their existing capabilities, market environment, and corporate strategy. Corporation's capability to learn emerging technologies and map changing industrial and market landscape would greatly help in the successful absorption of Industry 4.0 technology.

Proposition: 1

Successful Implementation and Absorption of Industry 4.0 Is Highly Dependent on the Dynamic Capabilities of the Corporation

In a highly competitive market, corporations would engage aggressively to seek leading positions. Industry 4.0 platform of technologies is a certain enabler in that direction. One of the reasons why Silicon Valley scores highly on creating and adopting technologies is that they have an easier access to highly trained human resources. One needs highly valued and qualified skilled human resources to drive, produce, and maintain the Industry 4.0 systems (Gatewood, Shaver, & Gartner, 1995). Corporations that are able to secure trained human resources would be better positioned in the competitive landscape. Thus, the successful implementation and absorption of Industry 4.0 platform of technologies are highly dependent on the human resources. The industry and governments need to work together to create human resources flexible for Industry 4.0 standards and requirements.

Proposition: 2

Successful Implementation and Absorption of Industry 4.0 Is Highly Dependent on the Human Resources of the Corporation

The Upper Echelon Theory clearly says that the success of an organization is highly dependent on the cognitive and leadership capabilities of the top management teams (Carmeli, Schaubroeck, & Tishler, 2011; Mueller & Barker Iii, 1997). The more acceptable and adaptable the top management teams are toward implementing emerging technological advances, the probability to succeed in accepting the change would greatly increase. Thus, the successful implementation of Industry 4.0 platform of technologies is not only highly dependent on the corporate strategy and evolving market dynamics but also on the cognitive and leadership capabilities of the top management. For example, Kodak was once a market leader but soon found that its leaders could not adopt the existing business towards the changing technological landscape. Subsequently, the board had to reconfigure the top management team from outside the group (Gavetti, Henderson, & Giorgi, 2003). The mélange of new culture brought by the new leadership and the existing work culture among the incumbent managers created tensions within the organization. Eventually, Kodak lost its market share and position.

Proposition: 3

Successful Implementation and Absorption of Industry 4.0 Is Highly Dependent on the Cognitive and Leadership Ability of the Top Management Team

The market width and depth greatly influence the corporations' ability to learn from multiple (strong and weak networks) sources and hedge market risks. The weak ties framework suggests that corporations which capitalize their discerning knowledge sources faster tend to learn and adapt faster (Granovetter, 1983; Liu & Duff, 1972). They also tend to manage uncertainty and risks faster. The weak ties help corporations to identify the opportunities, resources, and expectations much earlier. In order to forge weak ties, they must have diversified portfolios and ability to forge asymmetric complementary ties at different levels. Both market depth within a given service/product point and market width across various domains may help them to identify opportunities early. Similarly, the breadth and depth of market access

help them to manage the market risks and uncertainty better. Using similar logic, Industry 4.0 requires knowledge and resources from various domains including IoT, robotics, IT, manufacturing and blockchain. Corporations, which have the ability to identify the emerging technologies and opportunities early from the various discerning market sources, tend to learn faster and compete better. The Samsung Group is a highly diversified group operating in traditional logistics business as well as cutting-edge mobile technology business. Until 2008, Nokia (focused mobile technology firm) was a market leader and Samsung was a distant fourth. However, by 2012, Samsung was able to take over the market share of Nokia. Among the many reasons cited, diversified market and product strategy helped Samsung the most to outcompete Nokia.

Proposition: 4

Successful Implementation and Absorption of Industry 4.0 Is Highly Dependent on the Market Width and Market Depth of the Corporation

While conducting a cost-benefit analysis, if the cost of incorporating the Industry 4.0 technologies turns out to be more than the potential increase in overall benefits to the corporation, then, the corporation may engage with industry 4.0 service providers. Hence, it is best for the corporations to observe, learn, and outsource their requirements than to develop in-house. If the asset specificity of Industry 4.0 technology is very high such that the capital expenditure and market risk outweighs the potential benefits, then the corporation should engage with Industry 4.0 service providers instead of buying and installing them within their own factory premises (Hendrikse & Veerman, 2001; Susarla, Barua, & Whinston, 2009). The cost and organizational dynamics of outsourcing the Industry 4.0 capabilities may have an entirely different effect on the corporate strategy (Englander, 1988; Grover & Malhotra, 2003). The risks of outsourcing the capabilities to industry 4.0 services providers may also be very high. While, the in-house development of industry 4.0 capabilities may significantly increase the costs, but, buying the industry 4.0 services may significantly weaken proprietary capabilities.

Proposition: 5

Successful Implementation and Absorption of Industry 4.0 Are Highly Dependent on the Asset Specificity, Frequency of Use, and Market Risk

Impact of Industry 4.0 on Corporate Performance

The product differentiation strategy of any corporation is a strategy where it sells diversified range of products. For example, Mercedes sells different range and quality of cars. Subsequently, the diversified range of products increases the production and the marketing costs for the corporation. The biggest difference in Industry 4.0 generation and its predecessors is that it provides a higher degree of control, customization, and decentralized collaboration during production. The larger impact of these capabilities would be customer-centric products at lower costs (Wee, D., Kelly, R., Cattel, J., & Breunig, M., 2015). Corporations would be able to produce high quality, highly customized products to customers at a lower cost. Using Industry 4.0 capabilities, Corporations like Mercedes would be able to produce multiple ranges of models at a far lower production costs. The lower cost of production for Corporations

Incorporating Industry 4.0 in Corporate Strategy

would directly impact its financial bottom-line and increase its capabilities to provide a diversified range of products, therefore positively impacting corporate performance (Porter, 1980b, 1980c, 1985). Adoption of Industry 4.0 platform of technologies clearly provides a strong competitive edge to corporations engaging in product differentiation strategies and diversified product portfolios. Therefore, the chapter concludes that the Industry 4.0 platform of technologies greatly enhances the corporate performance.

Proposition: 6

Industry 4.0 Will Increase the Corporate Performance if It Is Coupled With the Product Differentiation Strategy of the Corporation

If a corporation's initial core strategy is cost leadership strategy, which involves selling a large number of products at lower prices, in that case, it would make strategic sense for it to invest in building resources that increase the output at a reduced price (Porter, 1980b, 1980c, 1985). The increases in production efficiency and volume strategy have a very high connection with increased corporate performance. Industry 4.0 capabilities are positioned to greatly reduce production costs. Thus, it makes strategic dexterity for the corporations to invest in Industry 4.0 capabilities. Corporations engaging in volume strategy must engage in cost-benefit analysis while making decisions on Industry 4.0 platform of technologies.

Proposition: 7

Industry 4.0 Will Increase the Corporate Performance If It Is Coupled With the Cost Leadership Strategy of the Corporation

New product development requires design thinking, customer feedback, market study, and customer profiling. It also requires strong, flexible, and rapid prototyping capabilities such as 3D printing, cad-cam design, and flexible machining capabilities. These are some of the capabilities that Industry 4.0 introduces to the current production systems (Qin, Liu, & Grosvenor, 2016). Industry 4.0 significantly enhances the ability to launch a product by integrating the *design to prototype any product or service* through customer requirements, market research, and, consequently, reducing the product development life-cycle time and cost considerably (Qin, J., Liu, Y., & Grosvenor, R., 2016). Data has shown that when the corporations launch new products at a higher rate they tend to have stronger performance over competitors (Pujari, Wright, & Peattie, 2003). Increased product development will positively impact the corporate performance (Rubera, Chandrasekaran, & Ordanini, 2016). Hence, accordingly, the corporate performance shall increase if Industry 4.0 is strategically integrated with the new product development strategy of the corporation.

Proposition: 8

Industry 4.0 Will Increase the Corporate Performance if It Is Coupled With the New Product Development Strategy of the Corporation

The implementation of Industry 4.0 is positioned to massively increase the productivity and reduce the overall costs of the corporation (Wee, D., Kelly, R., Cattell, J., & Breunig, M, 2015). As the Corporation engages in more aggressive actions, its competitors would also engage in similar actions. Higher productivity in a competitive market will put pressure on the overall margins. Higher pressure on margins

will make everyone worse off. Consequently, corporations would be forced to aggressively engage in actions and strategies to explore and exploit newer markets (March, 1991; Rodan, 2005). The new market exploration and exploitation strategy of the corporation will have a positive impact on its performance. Hence, in the long run, if a corporation chooses to aggressively engage with the Industry 4.0 platform of technologies, it will also have to invest and couple Industry 4.0 with the new market development strategies. Failing to do so may increase the risks of underutilization of production systems or may increase the inventory costs.

Proposition: 9

Industries 4.0 Will Increase the Corporate Performance if It Is Coupled With the New Market Development Strategy of the Corporate

The prominent non-markets strategies of any Corporation include corporate social responsibility (CSR) expenditure, support to public projects and goods, social value creation through community engagement, investment in industry associations, strategic lobbying with the government, and strategically managing different social and political groups (de Figueiredo, 2010; Sandel, 2012). These strategic investments are executed to create trust with the public and the government. Industry 4.0 is there to increase the productivity at the employee and organizational levels. It does not share any common characteristic with the non-market strategies of the corporation. The non-market strategies have no commonalities with the Industry 4.0 technology. Consequently, incumbent corporations must not replace their focus on non-market strategies with an increased focus on Industry 4.0. Rather, the current media narrative against automation and digitalization, the corporations digitalizing must invest more on non-market strategies to manage their public relations.

Proposition: 10

Industry 4.0 Will Have No Impact on the Corporate Performance if It Is Coupled With the Non-Market Strategies of the Corporation

Industry 4.0 will drastically increase labor productivity. It will also decrease the cost of organizational control. Consequently, the increased productivity may lead to a decreased requirement of manpower. The decreased requirement of the workforce may lead to an increased possibility of large-scale layoffs. The layoffs may create stress in the markets. Increased productivity will also lead to commodification, reduced margins, and higher stress in the markets. Consequently, this will lead to social, political, and economic tensions in the business-social environment, which will result in adverse consequences for the corporation (Flanagan & O'Shaughnessy, 2005). The negative spillovers because of increased productivity will also lead to reduced fund allocations and reduced tax incentives from the government. It can also lead to an increased governmental scrutiny of the enterprise's activities. On the other hand, engaging the government and public about Industry 4.0 and its greater impact on market equilibrium may lessen these fears. Autonomous vehicles are one example of Industry 4.0 (Beede, Powers, & Ingram, 2017). It is a disruptive innovation but it also threatens millions of jobs (drivers and transporters) that are dependents on vehicles. The recent tests by Google and Uber have not only attracted interest from tech enthusiasts but also raised concerns about the future of jobs.

Proposition: 11

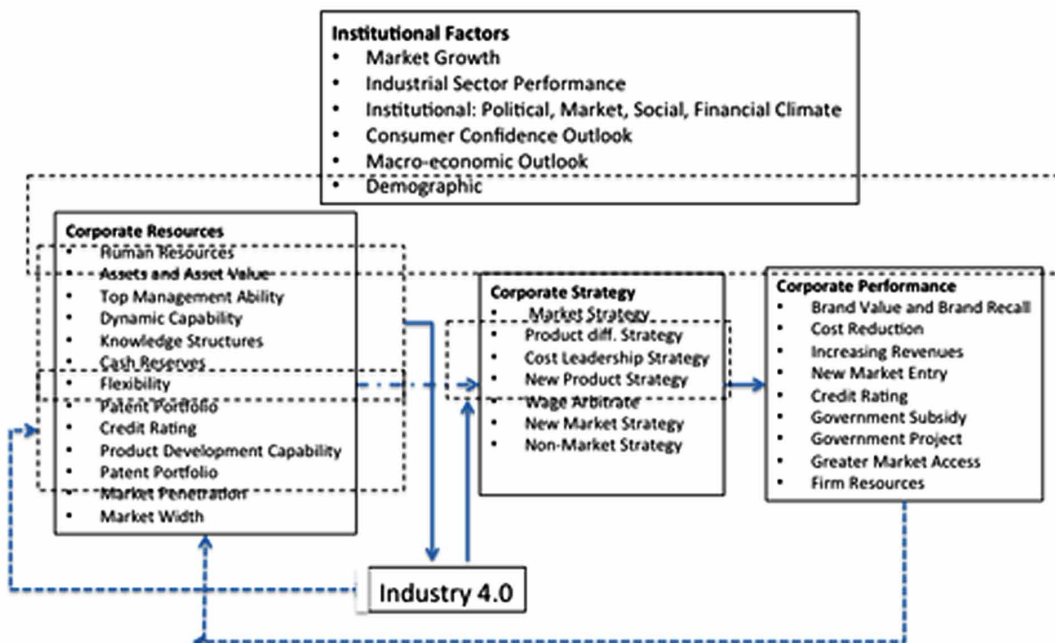
Industry 4.0 Implementation May Have a Negative Impact on the Corporate Performance if It Drastically Increases the Probability of Employee Layoffs or Market Competition

Using the 11 propositions developed in the chapter, the authors propose a diagram describing the position of Industry 4.0 within the corporate strategy landscape (Figure 3). The current available scholarship on Industry 4.0 is limited to conceptual definitions, adaptation in the industry, and operationalization within the manufacturing plant. The eleven propositions presented here, therefore, provide a perspective on how the corporate strategy and thereby corporate performance might get influenced while adopting Industry 4.0 enabling technologies. It also adds value to the current literature on Industry 4.0 by bringing in the strong corporate strategy perspective. The authors also hope to create a dialogue between Industry 4.0 solution providers/creators, corporations and policymakers.

CONCLUSION AND FUTURE OUTLOOK

This chapter looks not only at the changing technologies disrupting the current industrial sector but also at the corporate strategy and where and how Industry 4.0 adds value to the corporate performance. Corporations that learn and adapt faster tend to survive longer. They are generally more profitable while those, which are slow, tend to lose out to the competition. Industry 4.0 is an upcoming platform of technologies that is revolutionizing the rate of productivity per employee. It also has the promise to

Figure 3. Position of Industry 4.0 within corporate strategy landscape
Source: (Authors' Source)



dramatically reduce the cost of controlling and compliance incurred by corporations. Therefore, the future outlook of Industry 4.0 looks promising. However, governments and the civil society have yet to take into account the socio-political-economic spillovers. For instance, if the productivity of existing employees increases substantially with Industry 4.0, it would also require for the market demand to grow equally. In case of limited growth into the market demand, there will be a very high risk of an economic crash, market failure, and possible large-scale unemployment as was witnessed during the Great Depression in the 1930s and in 2007–2008. In order to manage uncertainty that might arise due to higher productivity and, therefore, unplanned layoffs, policymakers should study Industry 4.0 from different positions. They must study the consequential spillovers on the socio-economic equilibrium, labor productivity, market equilibrium, and the plan for any major market failure. They might also consider reorganizing schooling and training curriculums to help people secure employment.

This study is primarily at the organizational level and discusses the position of Industry 4.0 within the broader corporate strategy. However, the study found that there is very little critical literature on the in-depth research on the growth of Industry 4.0 in the long run from a public policy perspective. Industrial revolution 1.0 and onward have greatly changed the productivity of corporations and consequently enhanced their profitability. The strategic resources of the corporations changed dramatically post Industry 3.0. In this era, the intellectual resources of the firm were decoupled from manufacturing. Industry 4.0 will now further change the outlook of intellectual resources, manufacturing and supply chains. Industry 4.0 would further reduce human intervention and decrease costs of controlling and monitoring. Given the lowering of wages, the rising unemployment in northern countries, and rising wages in southern countries, multinational companies are already restructuring their supply chains and manufacturing strategies. However, it is still to be seen how these restructuring would impact the labor markets in North and South. Presently, China is the production engine of the world, which exports goods to the rest of the world at very competitive prices. Will Industry 4.0 change this trend? Similarly, as Industry 4.0 gains further acceptance, the labor force of the new industrial age would be highly productive and will have higher wages compared to the present worker, given our knowledge of the previous industrial revolutions. How will the rise in wages impact the current economic equilibrium at the country and international trade levels? As industries adopt Industry 4.0 technology the uncertainties over employment, trade secrets, and core strength continue to persist. The research calls for further studies on Industry 4.0 and its impact on outsourcing. In order for the corporation to successfully absorb Industry 4.0 technology, they also need a highly skilled workforce. The research calls policymakers to relook at the technical education to incorporate the requirements of the new industrial revolution.

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KEY TERMS AND DEFINITIONS

Corporate Competitiveness: Corporate competitiveness is the ability of the business to provide products and services as or more effectively and efficiently than relevant competitors.

Corporate Performance: Corporate performance is how effectively and efficiently a given business organization is achieving the objectives written in the strategy document of the business.

Corporate Productivity: Productivity is a measure used in many contexts. In this chapter, productivity is related to business performance. It is the ratio of output to input used in the production process.

Corporate Strategy: Corporate strategy is a set of statements that guides the corporation's decisions and actions to ensure its competitive position in the given industry and market.

Industry 4.0: Industry 4.0 is also known as “Smart Factory.” It is a platform for increasing the productivity by systematically using multiple innovations in data science, automation, IoT, big data, robotics, modular manufacturing, and decentralized decision making by machines.

Institutional Factors: Institutional factors are social structures (rules, culture, routines, norms, and conditions) formed by the government, markets, and society that influence the organizational behavior and functioning.

Labor Market: Labor market refers to the supply and demand for labor in which employees provide the supply and employers the demand. It is a major component of any economy and is intricately tied in with markets for capital, goods, and services.

Labor Productivity: Labor productivity is equal to the ration between a measure of output volume (gross value added) and a measure of input use (total number of hours worked).

Market Forces: Market forces are the factors that influence the price and availability of goods and services in a market economy. Market forces push prices up when supply declines and demand rises, and drive them down when supply grows and demand contracts.

Market Uncertainty: Market uncertainty is the situation where the current state of knowledge is such that (a) the order and nature of things are unknown, (b) the consequences, extent, or magnitude of circumstances, conditions, or events is unpredictable, and (c) credible probabilities to possible outcomes cannot be assigned.